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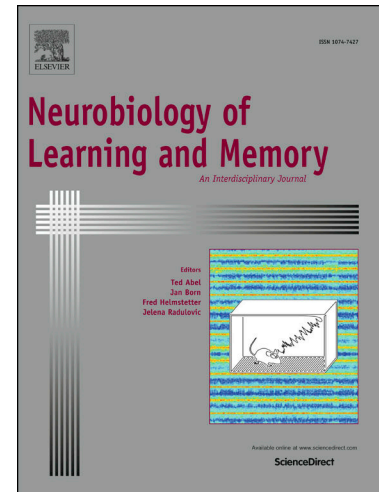
Editorial

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The Rescorla-Wagner Model: It Is Not What You Think It Is

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1. Lost in Translation

The Rescorla-Wagner model (henceforth abbreviated as RW (Rescorla & Wagner, 1972; Wagner & Rescorla, 1972) is widely recognized as the most famous theoretical model in Psychology, a claim that is backed by its ~12,001 citations to date. The influence of the model has been discussed at length in several reviews since its publication (Miller et al., 1995; Siegel & Allan, 1996; Walkenbach & Haddad, 1980), including Soto's in the current issue (Soto et al., 2023) who offer a historical perspective, tracing the model's pervasive influence across fields from cognitive science to neuroscience. Within learning theory, suffice it to say that RW has been referred to as "standard associative theory" (Hall, 1991), and subsequent formal theories of conditioning can grosso modo be classified either as explicit rivals (Gibbon & Balsam, 1981; Mackintosh, 1975; Miller & Matzel, 1988; Pearce, 1994; Pearce & Hall, 1980) or elaborations (e.g., (Honey et al., 2020; McLaren & Mackintosh, 2002; Sutton & Barto, 1990; Wagner, 1981). In behavioral neuroscience, RW and its temporal difference extensions (Sutton & Barto, 1990) showcase how theory-driven research can catalyze scientific progress. Applications of RW span across Psychology's rich tapestry of subdisciplines, including cognitive science, developmental, social (see Soto et al., 2023) and evolutionary psychology (Trimmer et al., 2012). In machine learning, RW's footprint can be seen in connectionist (Rumelhart et al., 1986) reinforcement learning (Sutton, 1988; Sutton & Barto, 1990) and Bayesian (e.g., Kang et al., 2023 in the current issue) approaches. These approaches have in turn fed new insights back into learning theory, cognitive science, and computational neuroscience. Remarkably, through its meteoric dissemination, RW has remained a reminder of the qualities that render any scientific theory useful—parsimony and falsifiability.

However, the model's cross-disciplinary success has come at a cost, both for RW and learning theory more broadly. For RW, this cost is manifested in widespread misconceptions about what the model can and cannot do. For learning theory, the cost has been having in RW not only its most eminent ambassador, but also the strawman whose perceived inadequacies are unfairly extrapolated to all associative theories. The goal of this Editorial is to dispel some of these misconceptions by providing historical context, emphasizing the model's conceptual innovations, and distinguishing between its perceived and actual limitations.

2. Conceptual Innovations

Unlike its reinforcement learning elaborations, RW was conceived as a model of Pavlovian conditioning, not instrumental learning, action selection or decision-making. The impetus for the model was to provide a single learning rule that would account for existing and, at the time, recently reported stimulus selection phenomena in Pavlovian learning such as the blocking, relative-validity and contingency effects (Kamin, 1968; Rescorla, 1968; Wagner et al., 1968). Reports of such effects in the late sixties had delivered a collective deadly blow to continuity theories that assumed that spatiotemporal contiguity (i.e., proximity) between the conditioned and unconditioned stimuli is a sufficient condition for learning. RW provided an elegant, mechanistic solution to the problem of selective stimulus learning that was a continuation of the Bush-Mosteller model (Bush & Mosteller, 1951). The latter model posits that changes in response probability on a given learning trial depend on the discrepancy between the asymptotic probability of responding (e.g., 1 if the stimulus is reinforced and 0 if it is not), and the current probability of responding, times a learning rate parameter. While the Bush-Mosteller model focused on observable responses, RW introduced the construct of associative strength (V) as the locus of learning. This construct refers to the strength of the connection between internal event

representations, and has been used since as a proxy for outcome expectancy. The learning rule then was defined as the discrepancy between current and asymptotic associative strengths, with asymptotic associative strength (λ) set by the intensity or magnitude of the US. This shift enabled RW to formalize the concept of prediction error as the difference between the predicted and actual outcome—a notion that has had a profound influence on subsequent learning models (see above) and neurobiological research (e.g., Iordanova et al., 2006; McNally et al., 2004; Roesch et al., 2007; Schultz et al., 1997).

Importantly, the prediction error in RW is an aggregate one (also termed summed, global, or common prediction error). In this framework, the prediction error is not based solely on the associative strength of the cue in question; rather, it is computed as the sum of the associative strengths of all cues present on a given trial. This assumption has become so deeply ingrained that, outside learning theory, the term "prediction error" often implies "aggregate prediction error" by default. This aggregation has two major consequences. First, it introduces the concept of cue competition, where cues compete for a finite amount of associative strength or predictive relevance. Unlike previous continuity models, RW posits that a cue's associative strength depends not only on its contiguity with the US but also on the presence of other cues. This enables the model to capture the relative, rather than absolute, informational value of each stimulus, allowing RW to explain cue competition phenomena, such as blocking, overshadowing, and relative validity effects, as well as contingency and US-preexposure effects (with the assumption that context can also acquire associative strength). Quigley and Haselgrove (Quigley & Haselgrove, 2023), in this special issue, extend the notion that cues compete for predictive power to outcomes, providing evidence for outcome overshadowing but not outcome relative validity effects.

The second critical consequence of an aggregate prediction error is that cues can acquire negative associative strength, providing a mathematical foundation for the concept of conditioned inhibition. Negative associative strength explains why a conditioned inhibitor reduces the conditioned response (CR) evoked by an excitatory cue in a summation test (by canceling out the excitatory cue's associative strength) and why a conditioned inhibitor displays retarded conditioning if later paired with a US (as its associative strength must shift from negative to positive). These predictions are difficult to account for under the Bush-Mosteller framework, which treats conditioned inhibitors as neutral cues with no inherent associative strength. While evidence of negative associative strength has been observed in the midbrain dopaminergic system, it is unlikely to be the sole neural mechanism for conditioned inhibition as reported by Esber et al., (Esber et al., 2024) in this issue. Specifically, these authors challenge the idea that the OFC encodes value per se, and argue that OFC activity reflects a model of the environment's associative structure. Other research on conditioned inhibition by Krueger et al., (Krueger et al., 2024) in this special issue provided evidence for sex differences with male rats showing significant inhibition of fear responses, whereas females displaying conditioned inhibition only in reward settings. This study underscores the need for a nuanced application of associative learning models, accounting for sex-specific variations in conditioned responses, particularly in fear-related contexts. Conditioned inhibition is increasingly recognized as a complex, multifaceted phenomenon, with evidence suggesting that inhibitory cues may evoke responses opposite to those elicited by excitatory cues in bidirectional response systems.

Yau and McNally (Yau & McNally, 2023), in the present special issue, provide evidence that the acquisition and extinction of fear learning is determined by a summed prediction error and that the variations in the effectiveness of the outcome to reinforce learning as determined by its predictiveness is reflected neurally at the level of opioid signalling in the ventrolateral periaqueductal gray. Relatedly to Esber et al., (2024) however, Yau and McNally (2024) challenge the symmetry between positive and negative error as stipulated by RW.

Unlike its predecessors, RW does not specify a strict rule for translating associative strength (V) into behavior, opting instead for a general assumption of a monotonic relationship between V and the likelihood or intensity of the conditioned response. Ghirlanda and Enquist (Ghirlanda & Enquist, 2023) in the present issue address the longstanding challenge of linking associative strength with observable behavior. Proposing a model that uses Poisson processes to represent behavior frequency, they provide a quantitative framework that strengthens the Rescorla-Wagner model's predictive utility. This approach allows for a more direct connection between learned associations and specific behavioral patterns. Relatedly, Navarro et al., (Navarro et al., 2024) report that reinforcement can influence two different response patterns, namely, goal-tracking and sign-tracking behaviors using rat autoshaping procedures. They report that reinforcement primarily increases goal-tracking, while nonreinforcement decreases it. In complement to this research, Maria-Rios et al., (María-Ríos et al., 2023) also in this special issue, provide evidence that that goal-tracking is sensitive to blocking whereas sign-tracking is sensitive to devaluation. This study supports an extension of the Rescorla-Wagner model, known as HeiDI, which attributes different associative origins to these behaviors based on stimulus characteristics.

Notably, while the RW model describes the growth of associations (positive or negative) between internal representations of cues and outcomes, it does not specify the content of these associations. RW remains agnostic as to whether the cue links to sensory, motivational, or motor components of the outcome representation—or any combination of these. This agnosticism has led to misconceptions about RW's theoretical stance: contrary to some interpretations, RW is neither a strictly "model-free" theory (in which cues merely acquire affective value) nor a "model-based" theory (in which cues access detailed sensory information about future events). Furthermore, while RW's algorithm was originally formulated to operate at the level of entire stimulus representations, it is equally applicable to componential representations at more granular levels, such as individual features or elements of a stimulus (Blough, 1975; McLaren & Mackintosh, 2002).

The innovations introduced by the Rescorla-Wagner model have had lasting impacts across psychology, neuroscience, and beyond. By incorporating notions of prediction, expectancy, and aggregate prediction error, the model transcended simple Pavlovian conditioning to become a touchstone for cognitive theories of learning and a foundation for computational neuroscience models. For example, Kang et al. (Kang et al., 2024) expanded the Rescorla-Wagner model within a Bayesian framework, introducing uncertainty as a critical component of learning. By integrating elements such as the Kalman filter, the authors show how Bayesian approaches can generalize the model's applicability to varied conditioning scenarios. This work offers a computational perspective, presenting Bayesian reinforcement learning as a powerful complement to the classic Rescorla-Wagner framework.

3. Conceptual Misunderstandings

The greatest legacy of the RW model may be the extensive body of research and theoretical advancements it has inspired over the past several decades (see Soto et al., 2023 in the current issue). Notably, cue interaction phenomena such as overexpectation and superconditioning—now gaining attention in neuroscience—were first identified through theoretically derived predictions of the model. While RW's numerous validated predictions have reinforced its success and influence, it is often its perceived shortcomings that have driven the field forward. However, not all of these perceived limitations stem from accurate interpretations of the model. In this section, we address misunderstandings that arise from overly constrained or unnecessarily simplistic applications of the model.

The RW model is not inherently model-free. In their seminal papers (Rescorla & Wagner, 1972; Wagner & Rescorla, 1972), the authors were explicitly agnostic about the specific contents of associations, making RW compatible with both model-free and model-based frameworks. This theoretical neutrality in the original publication, coupled with its role in shaping explicitly model-free frameworks like temporal difference reinforcement learning (TDRL), contributed to the widespread yet erroneous belief that RW is exclusively model-free. Consequently, this misinterpretation has led to the assumption that empirical demonstrations of model-based learning challenge the RW model.

US devaluation studies. A notable example involves US devaluation studies. In such experiments, a cue initially conditioned to predict food is later tested after the food is paired with an aversive stimulus, such as gastric malaise. At test, the conditioned response to the cue typically decreases, reflecting the animal's sensitivity to the devalued outcome. If RW were strictly interpreted as encoding only an inflexible, cached value (a hallmark of model-free learning), it would indeed fail to predict devaluation effects. This is because updating the cue's value would require direct pairing with the devalued US, which does not occur during the devaluation procedure.

Unlike TDRL, however, the Rescorla-Wagner (RW) model focuses on the strength of associations between events. Through these learned associations, predictive cues can activate various components of an outcome's representation, including response, affective, and sensory elements, as highlighted by Rescorla (1973). Accordingly, RW can account for devaluation by proposing that the cue activates the sensory features of the food outcome (e.g., flavor, texture). During devaluation, these sensory features become associated with elements of gastric malaise. At test, presentation of the cue reactivates the sensory features of the food outcome, which in turn trigger the representation of gastric malaise, leading to a reduction in cue-evoked conditioned behavior. This representational flexibility demonstrates that RW is not inherently restricted to model-free interpretations but can also capture aspects of model-based learning when appropriate assumptions are applied.

Extinction following US devaluation. In a related line of work, Holland and Rescorla (Holland & Rescorla, 1975) reported that devaluing an outcome prior to extinction training protected its associated cue from extinction. That is, whereas repeated non-reinforced presentations of a CS paired with a high-value US led to a reduction in conditioned responding that persisted during recovery tests, devaluing the US after acquisition but before extinction attenuated extinction, revealed in a strong recovery of the conditioned response once the US regained its original value.

The authors compared these findings to the phenomenon of protection from extinction, first described by Chorazyna (Chorazyna, 1962), where extinction training conducted in the presence of a conditioned inhibitor attenuates extinction learning. This phenomenon is readily explained within the RW framework, as the conditioned inhibitor lowers the expectation of the outcome, thereby diminishing negative prediction errors. Similarly, Holland and Rescorla (1975) interpreted their findings by suggesting that the cue activates a devalued US representation, which, when compared to the omitted US, generates a smaller negative prediction error than a non-devalued US. As a result, little extinction learning would occur.

Learning about cues of low motivational significance. Another example of an overly constrained application of the RW model is sensory preconditioning, where two neutral cues are initially paired, and one is subsequently conditioned with a motivational outcome. At test, the previously unpaired cue elicits a response similar to that of the conditioned cue. Critics have argued that RW cannot account for sensory preconditioning because the asymptotic associative strength (λ) of neutral cues should be low or near zero (Miller et al., 1995). However, RW does not mandate a binary concept of reinforcing power, nor one that can only be applied to biological reinforcers. Instead, λ can be viewed as a continuum, influenced by factors like stimulus intensity and biological relevance. A mild stimulus (e.g., soft white noise) may carry a modest λ value, supporting learning and eliciting a mild orienting response, while a stronger stimulus might elicit more robust responses. Thus, neutral stimuli with nonzero reinforcing power can still participate in associative learning, as supported by evidence of cue competition even when both cues are neutral (e.g., Esmoris-Arranz et al., 1997).

One particularly noteworthy example of the flexibility of the RW model in accounting for learning, as described by Rescorla (Rescorla, 1972) is its application to second-order conditioning. Second-order conditioning is similar to sensory pre-conditioning in that it involves the pairing of two relatively neutral stimuli (e.g. a light and a tone); however, this pairing occurs after one of the stimuli has already been conditioned with an outcome of biological significance. As a result, second-order conditioning allows for excitatory associations to form between the two sensory stimuli, while simultaneously enabling inhibitory associations to develop between the second-order conditioned stimulus and the anticipated outcome. According to the Rescorla-Wagner (RW) model, these distinct forms of learning emerge because a positive prediction error is generated when the two sensory stimuli are paired, while a negative prediction error occurs when the anticipated outcome is absent.

Examples of facilitative cue interactions, such as potentiation and augmentation (cite), can also be explained within the RW framework (Pineño, 2007). In potentiation, training a target cue in compound with another one can result in greater responding than training it in isolation—the opposite observation to overshadowing. In augmentation, pretraining the accompanying cue before compound conditioning with the target cue results in greater responding to the latter than omitting the pretraining—the opposite observation to the blocking effect. RW can explain such examples of facilitation by assuming that cues presented in compound can also form associations (so called within-compound associations). Through such associations, a cue presented at test can activate a memory of the formerly accompanying cue, which in turn will activate the outcome representation. This indirect activation will support responding over and above the cue's own direct association with the outcome. Whether facilitation or competition will be seen thus

depends on the balance of various associations in play, all of which could be acquired via the RW rule.

Examples of facilitative cue interactions, such as potentiation (e.g., Galef & Osborne, 1978) and augmentation (e.g., Batson & Batsell, 2000), can also be accommodated within the Rescorla-Wagner (RW) framework (Pineño, 2007). Potentiation occurs when training a target cue in compound with another cue results in greater responding than training the target cue in isolation—an effect opposite to overshadowing. Similarly, augmentation refers to the observation that pretraining the accompanying cue before compound conditioning leads to greater responding to the target cue compared to when pretraining is omitted—an effect opposite to blocking. The RW model can account for such facilitative interactions by incorporating the concept of within-compound associations, whereby cues presented together form associations with each other (Luque et al., 2013). At test, a target cue can indirectly activate the representation of an accompanying cue via this within-compound association. The accompanying cue, in turn, activates the outcome representation, enhancing responding above what the target cue could elicit based on its direct association with the outcome alone. As a side note, this can explain why blocking is rarely complete. Whether facilitation or competition emerges ultimately depends on the interplay of multiple associations acquired during training, all of which are governed by the RW learning rule.

Restoration of conditioned responding following extinction. Perhaps the best-known limitation of RW is its inability to explain the persistence of associations following extinction, as evidenced by spontaneous recovery, renewal, and reinstatement effects. However, as Delamater and Westbrook (Delamater & Westbrook, 2014) have argued, extinction occurs within a context, and contextual cues are capable of contributing to the aggregate prediction error. Consequently, inhibitory learning from extinction trials is distributed across both the target cue and the context. This distributed inhibition protects the target cue from complete extinction, and if the cue is tested in a different context, the original conditioned response is partially restored (Chorazyna, 1962). Consistent with this perspective, extinction across multiple contexts leads to deeper extinction effects (e.g., Bustamante et al., 2024). RW further predicts that while the context can protect the cue from total extinction, it does not prevent some degree of associative loss (unlearning), consistent with the incomplete nature of spontaneous recovery and renewal effects.

In this special issue, Kennedy et al. (Kennedy et al., 2024) revisit the question of what is learned in extinction using a PIT procedure as employed previously by Delamater (Delamater, 1996, 2012; Delamater et al., 2017). The authors show that the strength of acquisition plays a role in the content of learning during extinction and that content can be influenced by presenting the CS in an explicitly unpaired fashion with the US rather than using the standard extinction protocol of repeated non-reinforcement.

Extinction of inhibition. Another notable limitation of RW is its prediction that presenting a conditioned inhibitor (a cue with negative associative strength) in isolation will extinguish its inhibitory properties—a prediction contradicted by the findings of (Zimmer-Hart & Rescorla, 1974). Sosa and Ramirez (Sosa & Ramírez, 2019) noted that most tests of this prediction have been conducted in excitatory contexts, which may reduce the prediction error by canceling out the inhibitor's negative strength. Although attempts to eliminate this confound have yielded

mixed results, it remains an area that warrants further investigation to clarify whether extinction of inhibition occurs under more controlled conditions.

These misunderstandings do not imply that all of RW's limitations can be resolved by reinterpretation. Indeed, some of the model's challenges have catalyzed the development of competing theories. For instance, RW's difficulty with conditional, nonlinear discriminations encouraged the development of configural models, which were then complemented by componential models (e.g., (Harris, 2006; McLaren & Mackintosh, 2000; McLaren & Mackintosh, 2002; Wagner, 2003). Similarly, the model's overly parsimonious assumption that the associability of cues is a constant solely determined by their salience, which is challenged by ample evidence of learning-related associability changes, led to the development of attentional theories of associative learning (Esber & Haselgrove, 2011; Mackintosh, 1975; Pearce & Hall, 1980). Furthermore, its stance on cue competition as a pure learning deficit led to the exploration of performance-deficit models (e.g., Miller & Matzel, 1988; Stout & Miller, 2007), while the absence of a timing mechanism spurred the creation of real-time models of associative learning within both animal learning (Gallistel & Gibbon, 2000; Gibbon & Balsam, 1981) and reinforcement learning domains (Sutton & Barto, 1990).

Through this discussion, we hope to clarify some of the more nuanced aspects of the Rescorla-Wagner model. While the model is not without its challenges, many of its perceived limitations stem from oversimplified applications rather than genuine theoretical flaws. By understanding these subtleties, researchers can better appreciate the model's robustness and the complexities involved in empirically testing any learning theory.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT-4 in order to improve readability of the text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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